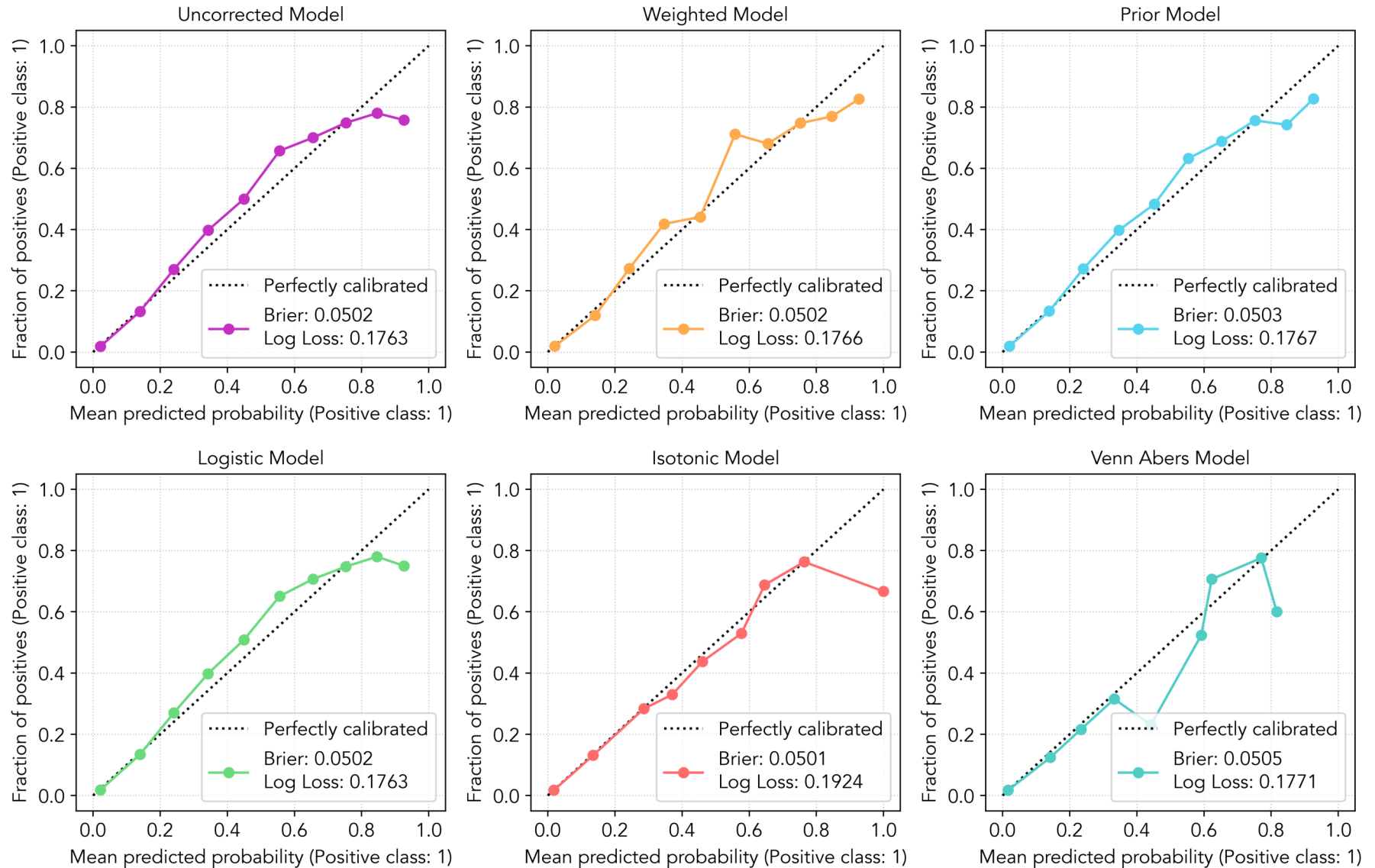


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



LogisticRegression - Calibration Curves

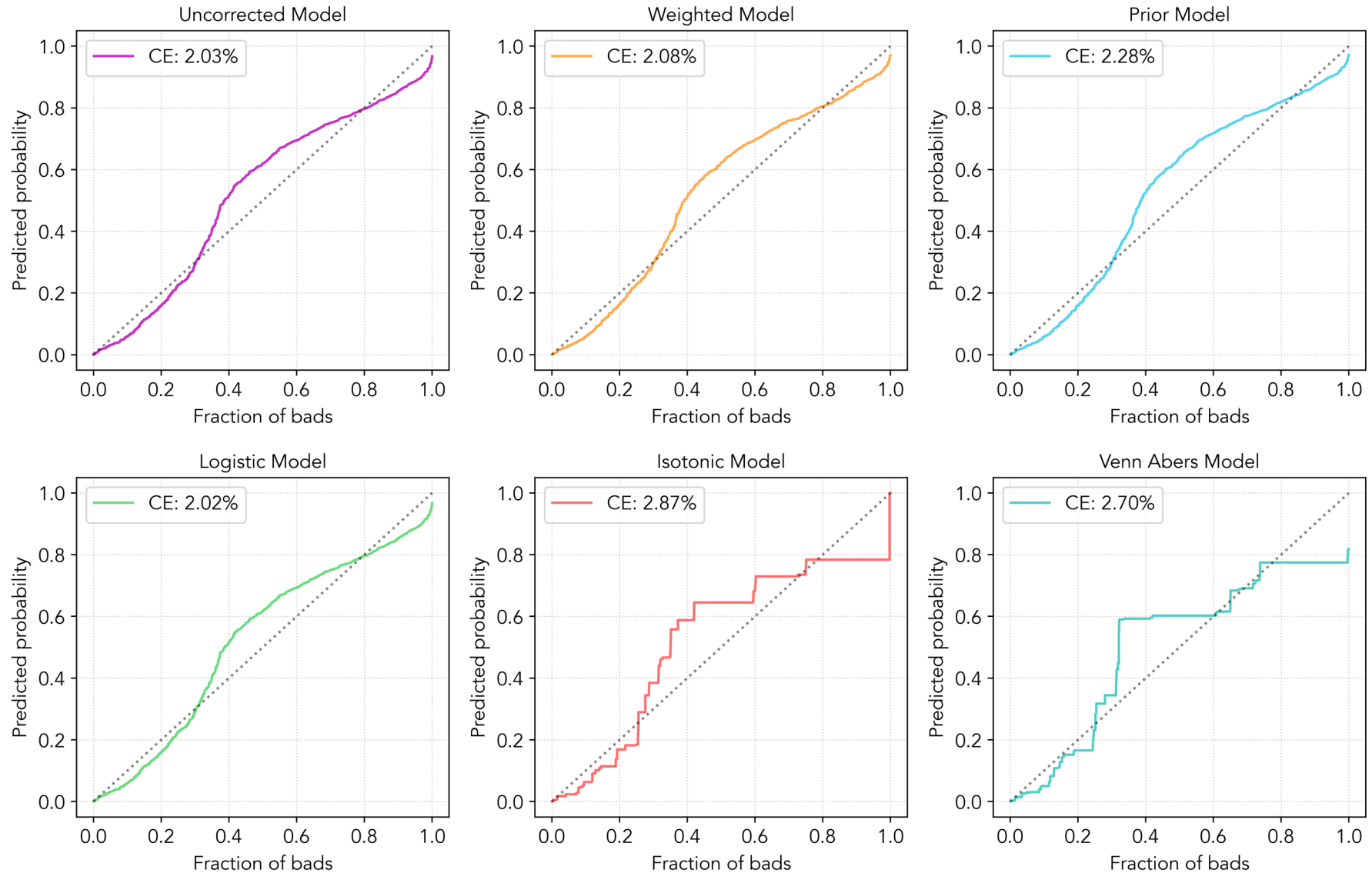


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



LogisticRegression - CCP Curves

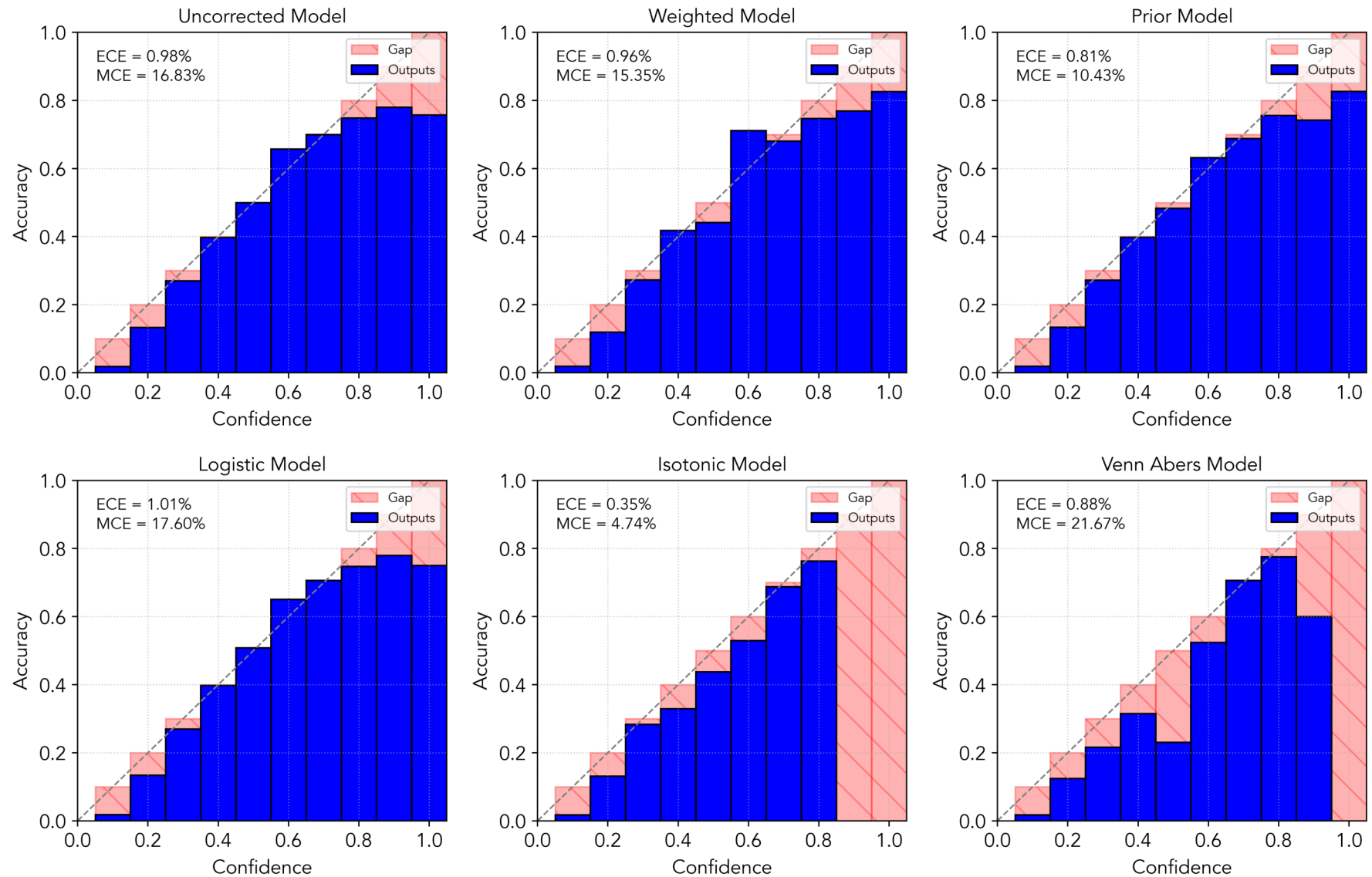


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



LogisticRegression - Reliability Diagrams

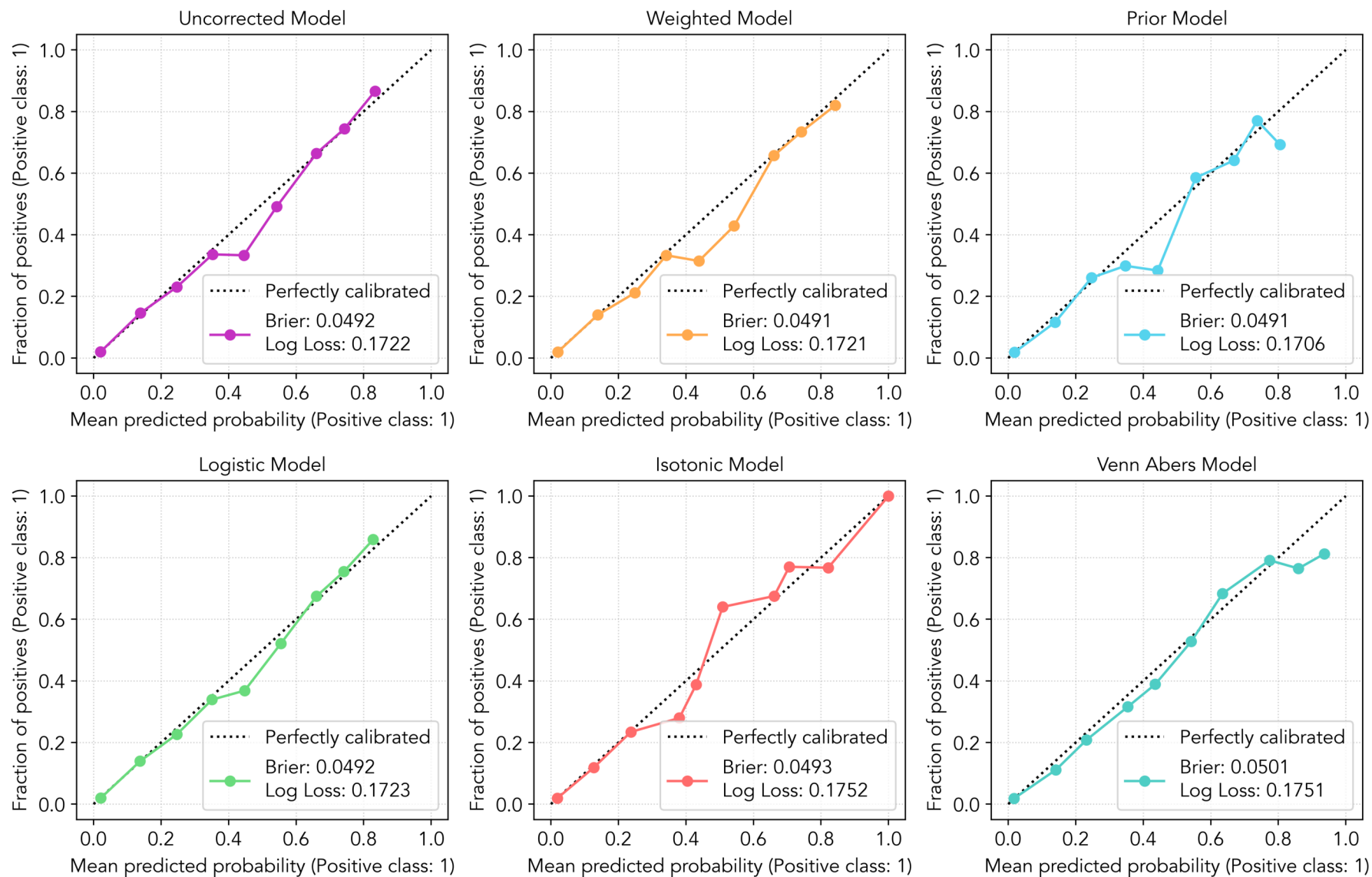


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



CatBoost - Calibration Curves

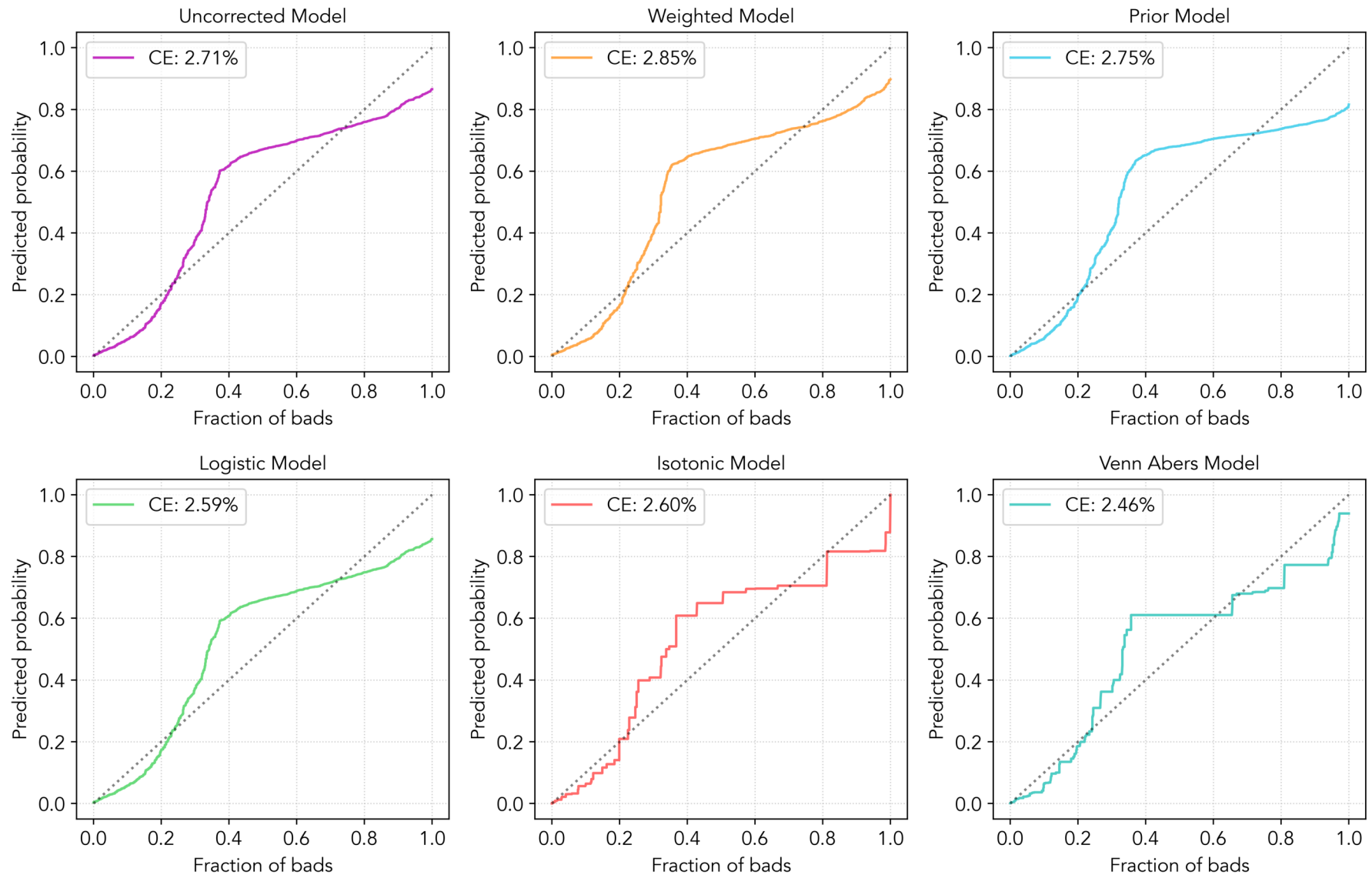


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



CatBoost - CCP Curves

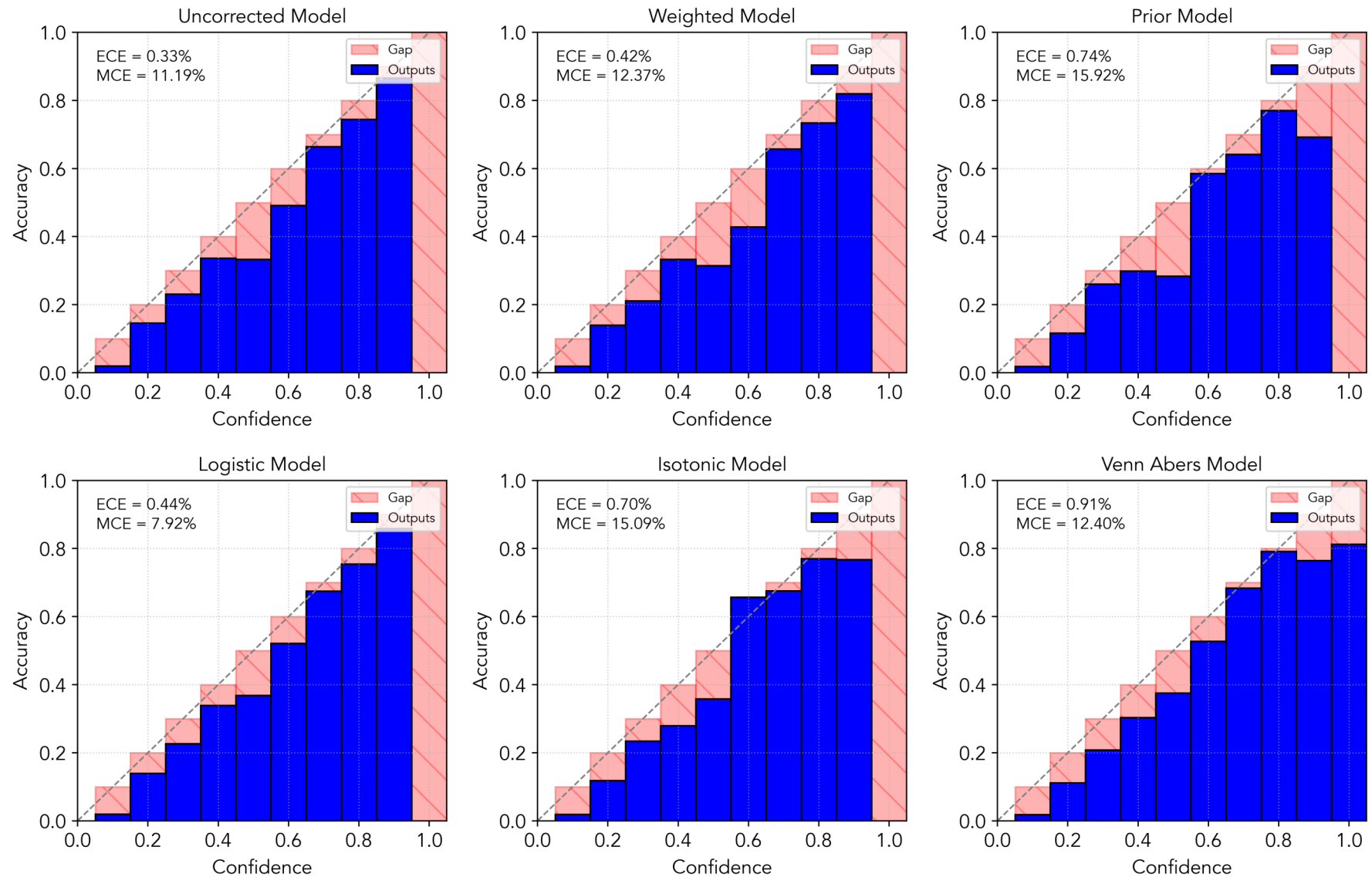


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



CatBoost - Reliability Diagrams

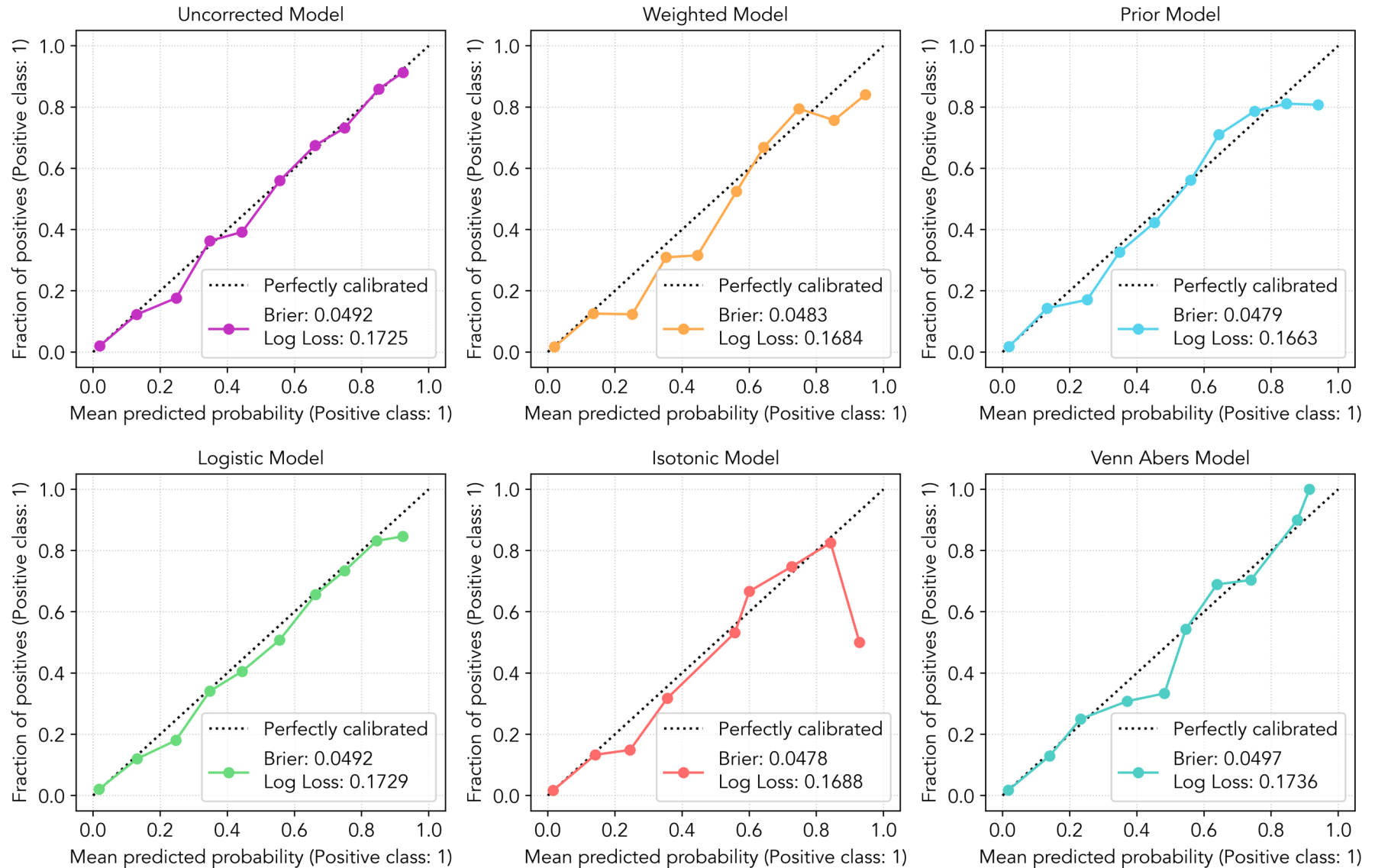


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBoost - Calibration Curves

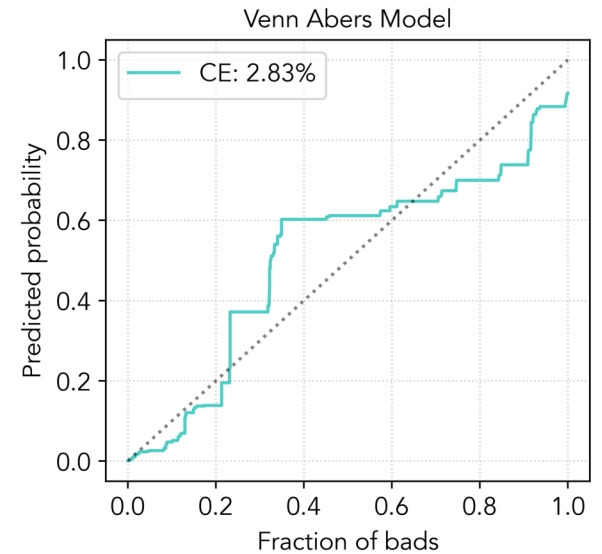
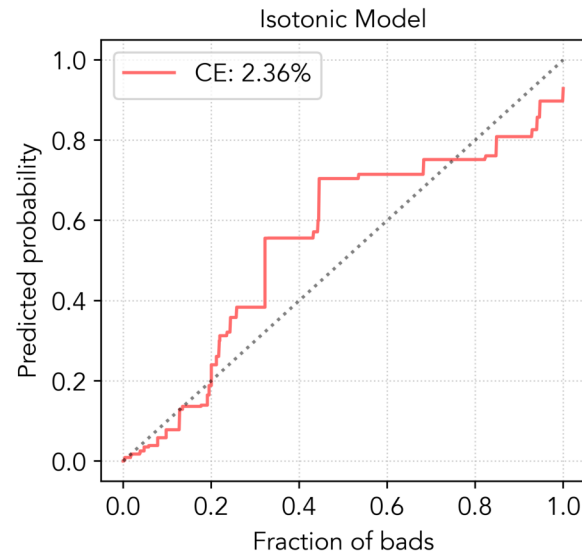
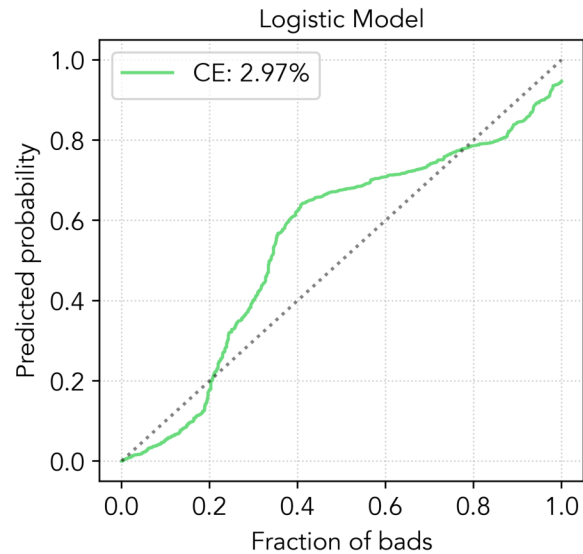
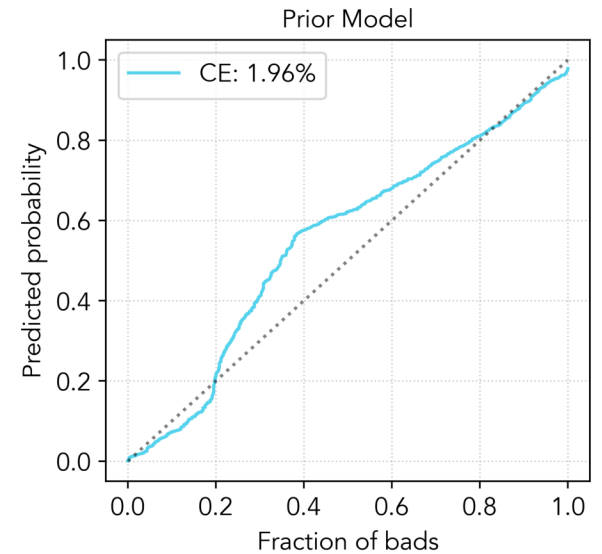
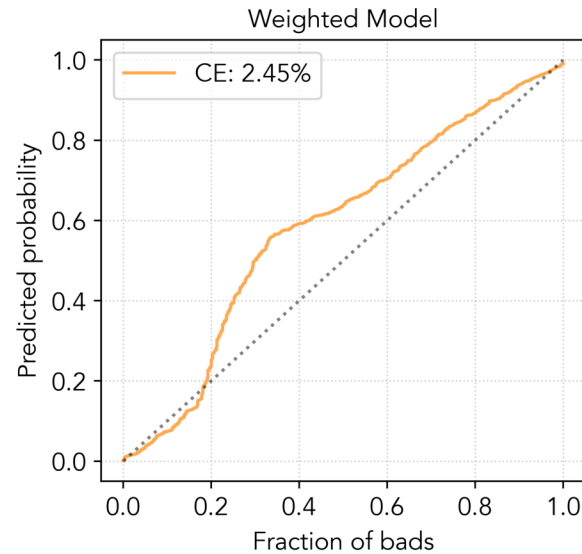
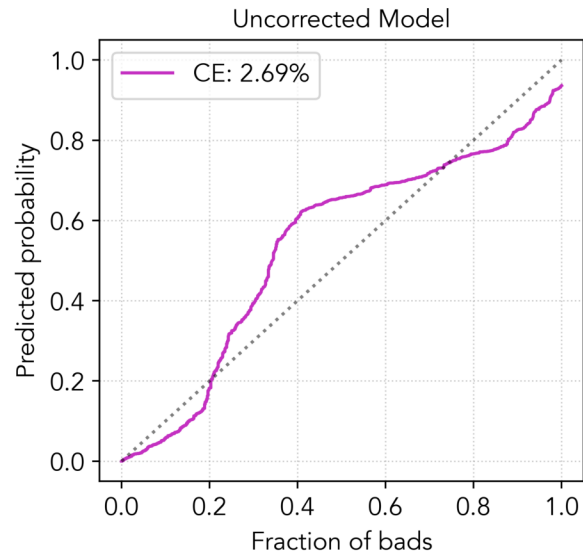


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBoost - CCP Curves

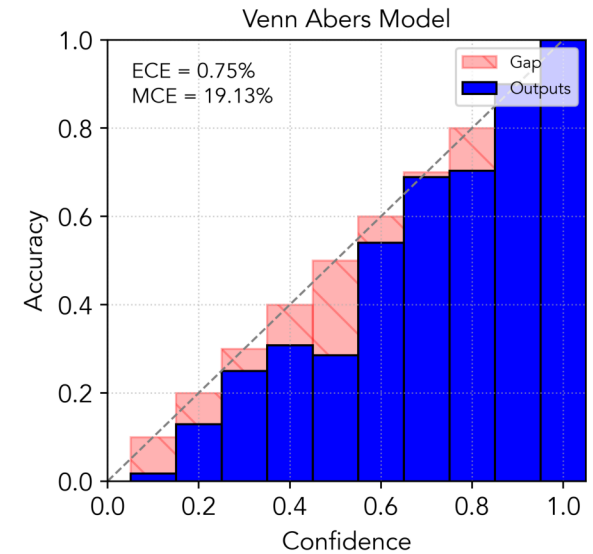
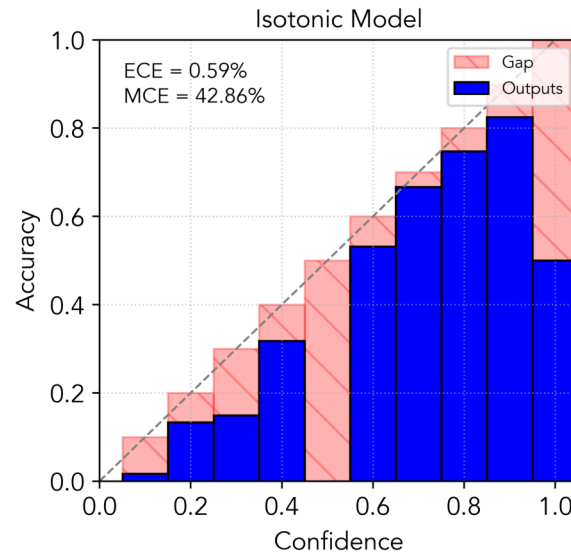
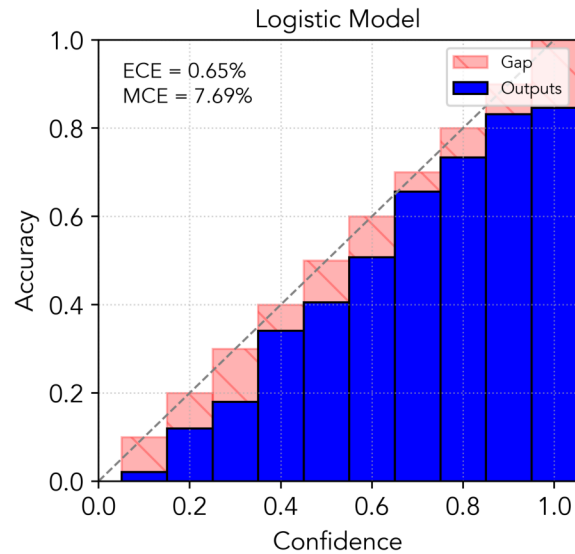
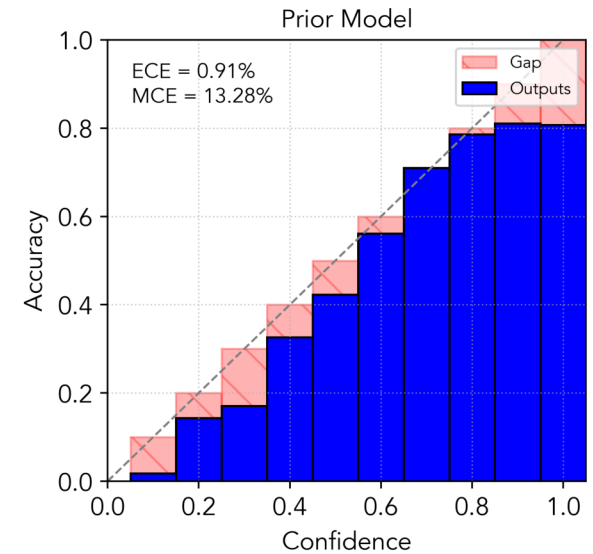
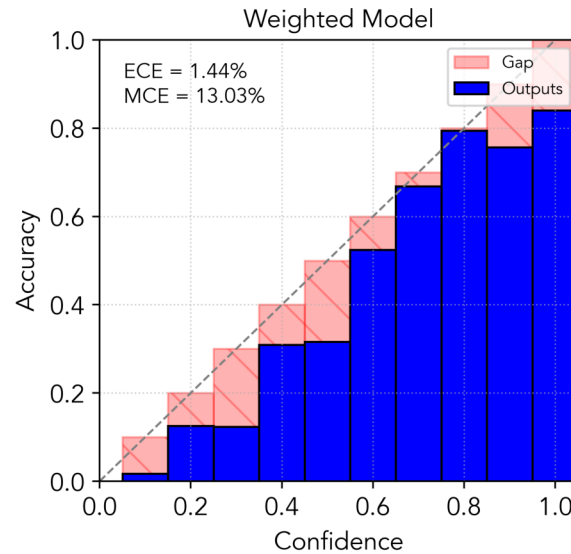
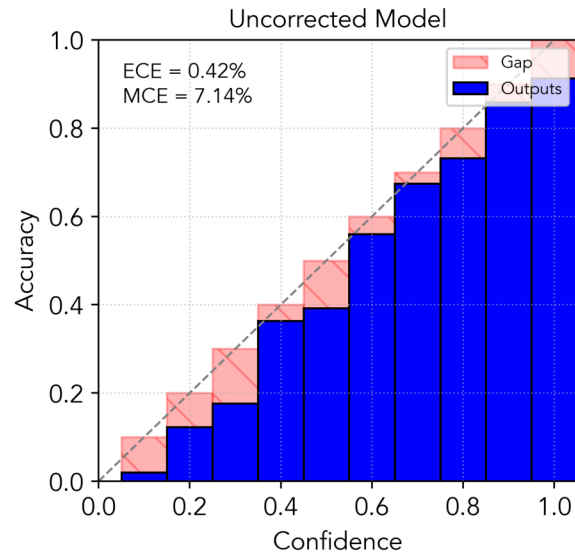


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBoost - Reliability Diagrams

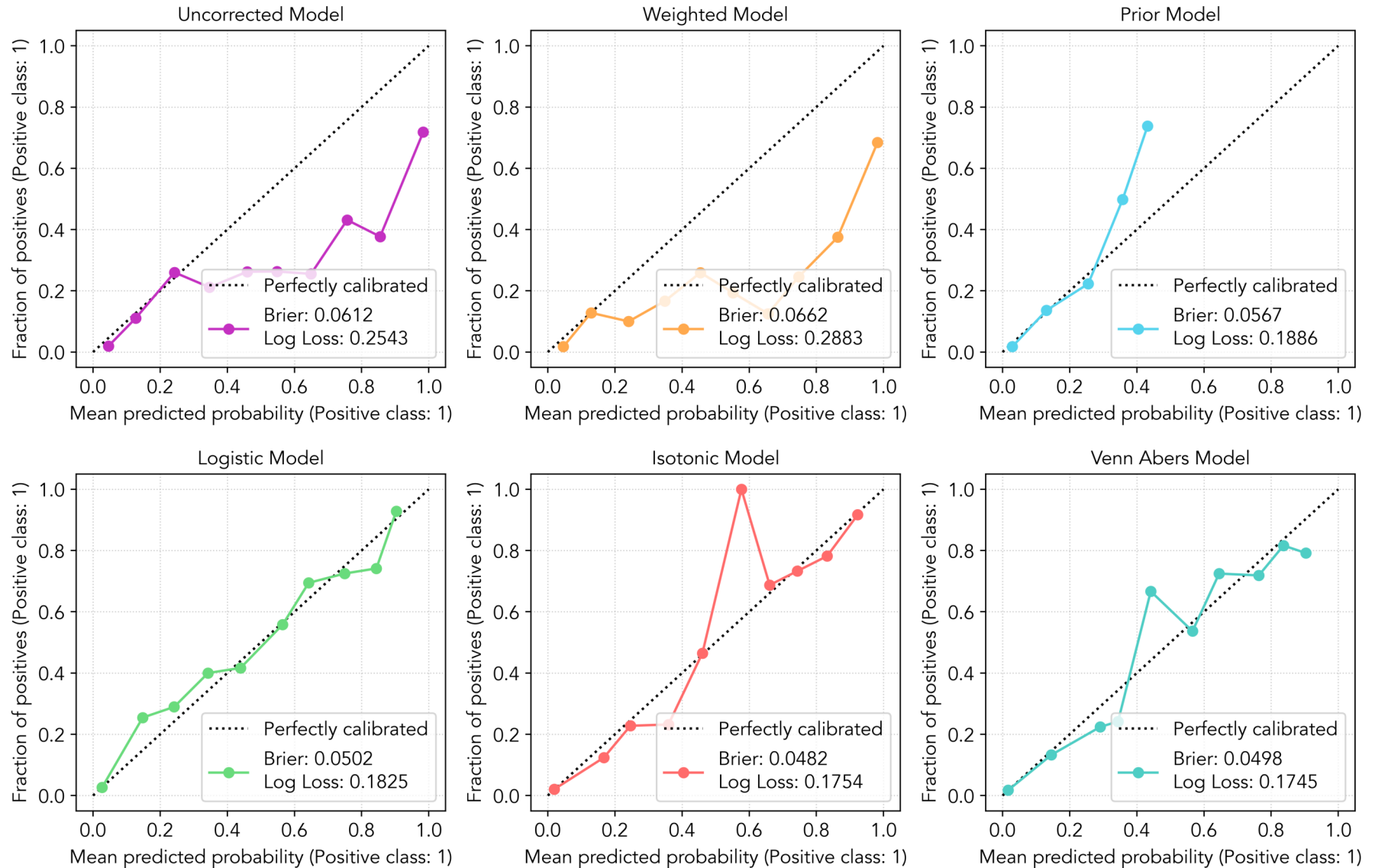


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBRandomForest - Calibration Curves

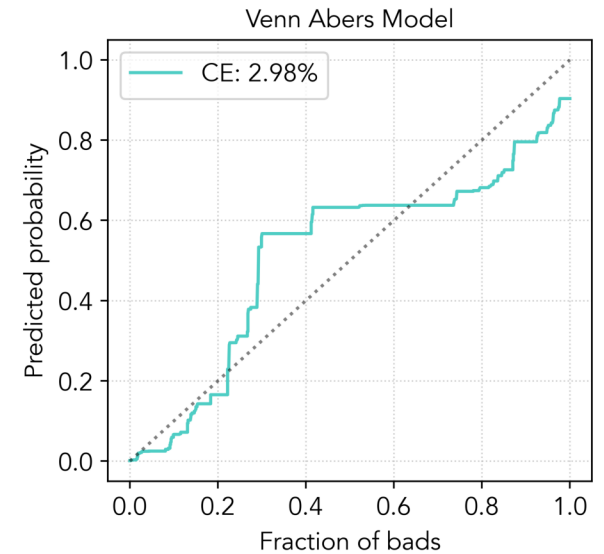
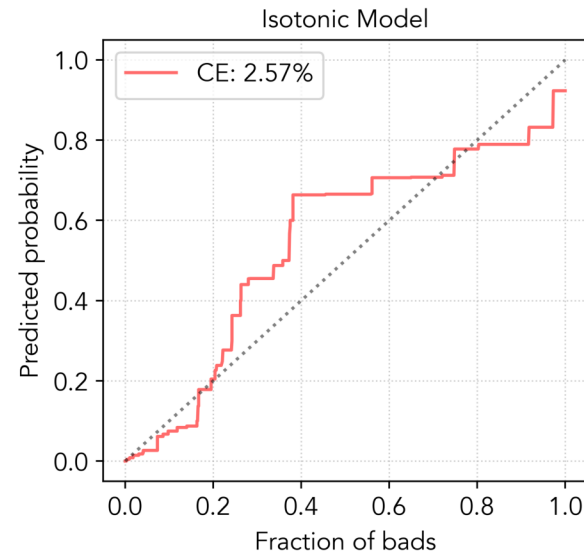
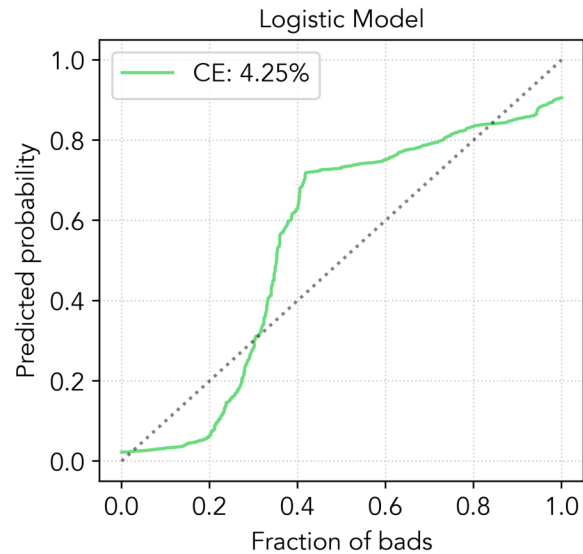
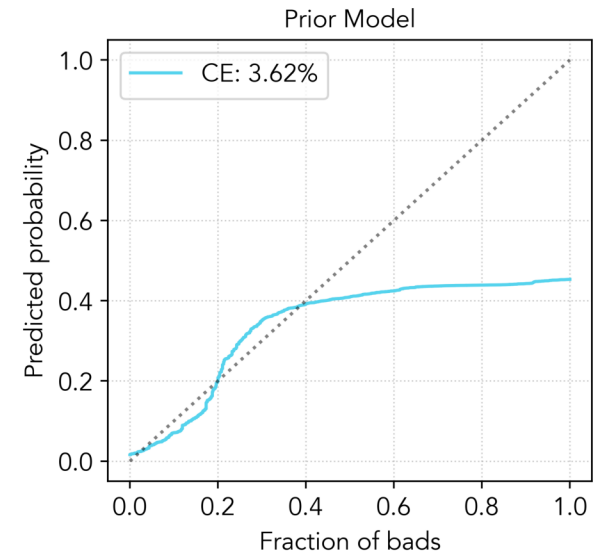
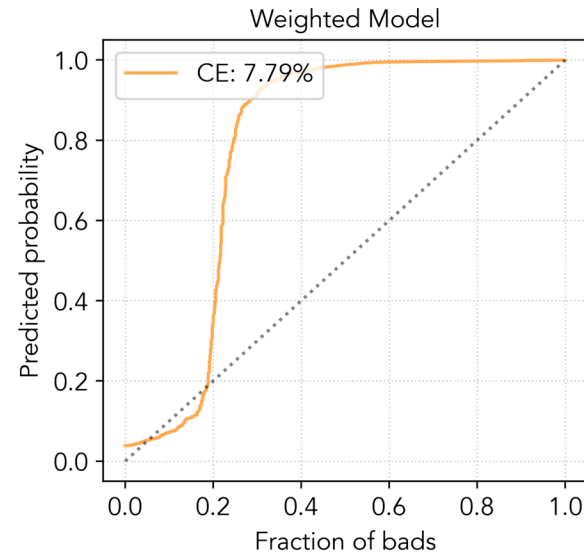
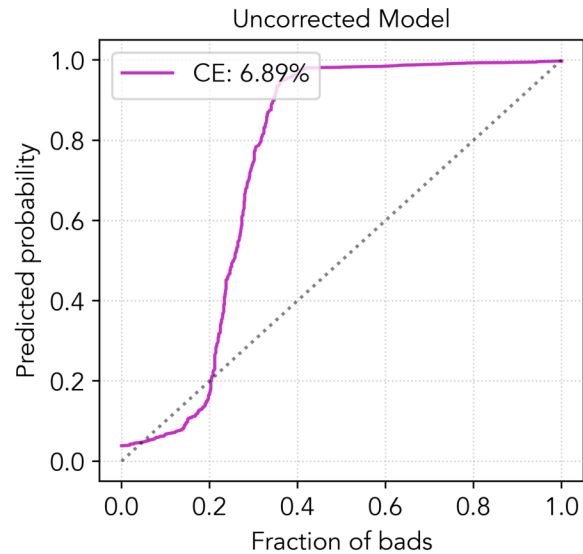


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBRandomForest - CCP Curves

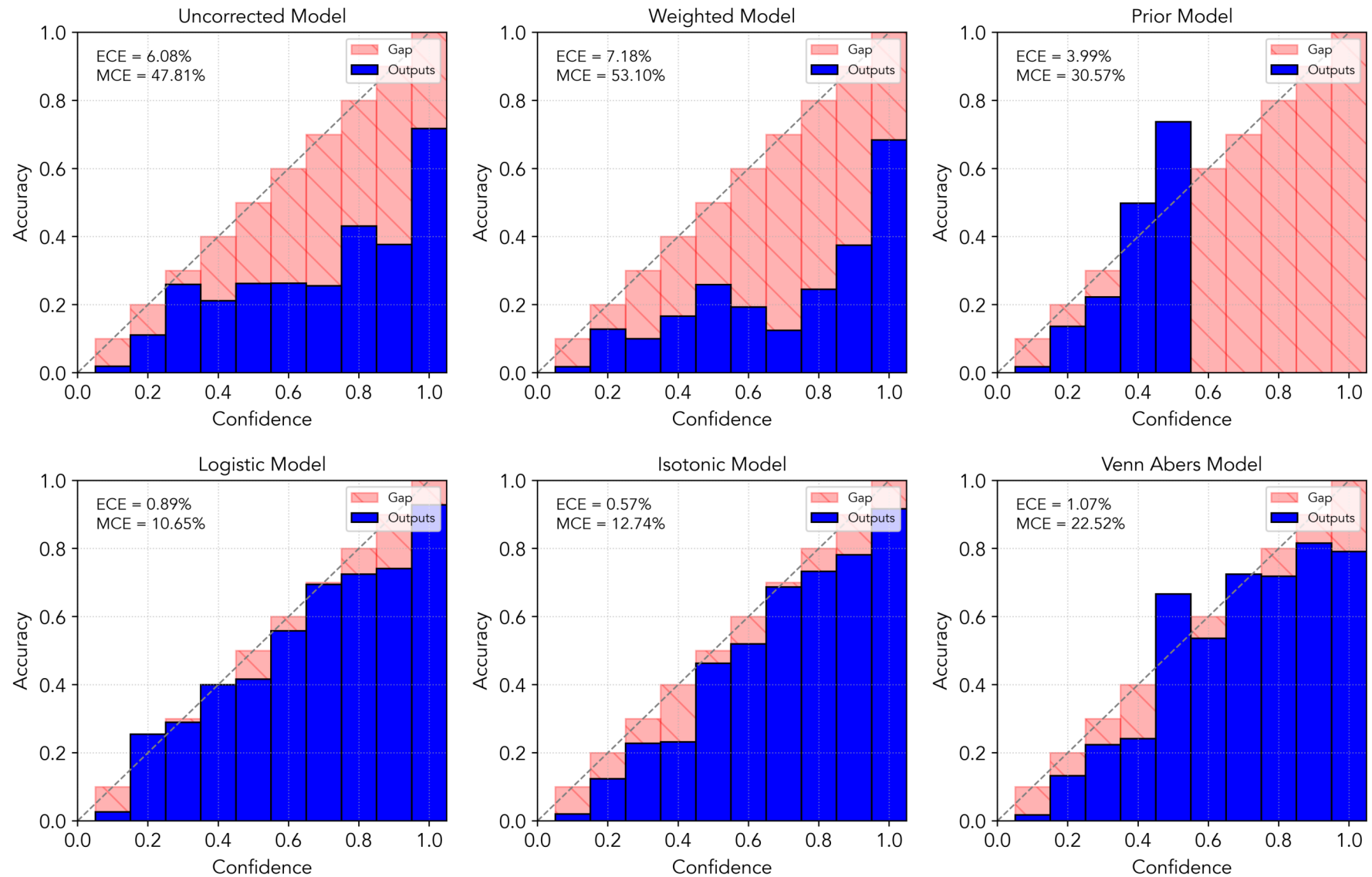


Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



XGBRandomForest - Reliability Diagrams



Evaluating Calibration Accuracy

Calibration metrics after undersampling with 50/50 imbalance ratio



PROBLEM STATEMENT

- When training on balanced samples (50/50) for rare events, predictions must be corrected to population prevalence p_t .

DATA & SPLIT

- Real banking dataset ($\approx 10\%$ default rate); stratified 70/30 train/test.
- Balanced training sample via undersampling to 50/50.

CALIBRATION VARIANTS

- Uncorrected: train on original data.
- Weighted (King-Zeng): class/sample weights:

$$w_0 = \frac{1-p_t}{1-p_s}, \quad w_1 = \frac{p_t}{p_s}$$

- Prior-corrected (King-Zeng §4.1):

$$\beta_0^{\text{corr}} = \beta_0^{\text{orig}} - \log\left(\frac{1-p_t}{p_t} \frac{p_s}{1-p_s}\right)$$

CALIBRATION VARIANTS (CONT.)

- Logistic recalibration (Platt):
 $\hat{p} = \sigma(az + b), \quad \sigma(x) = \frac{1}{1+e^{-x}}$
- Isotonic calibration: non-parametric mapping with CalibratedClassifierCV (prefit).
- Venn-Abers: inductive split, probabilistic intervals.

EVALUATION METRICS

- Metrics: Brier, Log loss, AUC; ECE, MCE & CE.

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2$$

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$$

$$\text{MCE} = \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$$

$$\text{CE} = \frac{1}{n} \sum_{i=1}^n |\hat{p}_i - y_i|$$